More pytorch

why Autograd?

(1)

$y = x^2$

$\frac{dy}{dx} = 2x$

$x = 1 = 2$

$x = 2 \quad , \quad 4$

c.g

(2)

$y = x^2$

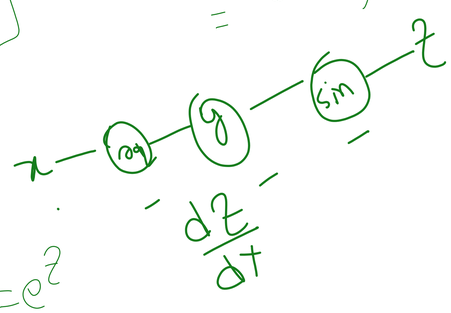
$z = \sin(y)$

$\frac{dz}{dx}$

$\frac{dz}{dy} \times \frac{dy}{dx}$

$= \cos(y) \times 2x$

$\Rightarrow \boxed{2x \times \cos(x^2)}$



(3)

$y = x^2 \quad z = \sin y \quad u = e^z$

$\frac{du}{dx} = \frac{du}{dz} \times \frac{dz}{dy} \times \frac{dy}{dx}$

[when complexity arising
code becomes complex]

Computational graph :

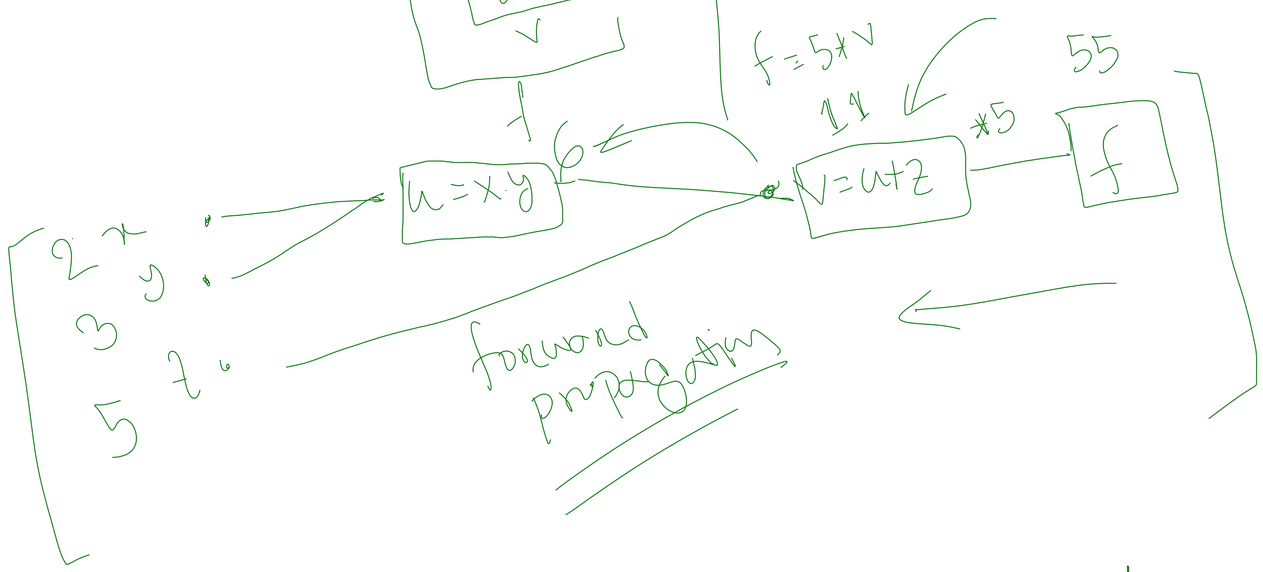
DPW

$f(x,y,z) = 5(xy+z)$

$u = xy$

$v = u+z$

$f = 5*v$



$\frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial x}$

#chain rule

$a - b - c$

$fun(a)$ $fun(b)$

$b = a^2$ $c = 10b$

$c = 10a^2$

$= 10b$

10

$\cancel{10b}$

$2a$

$2a$

$2a$

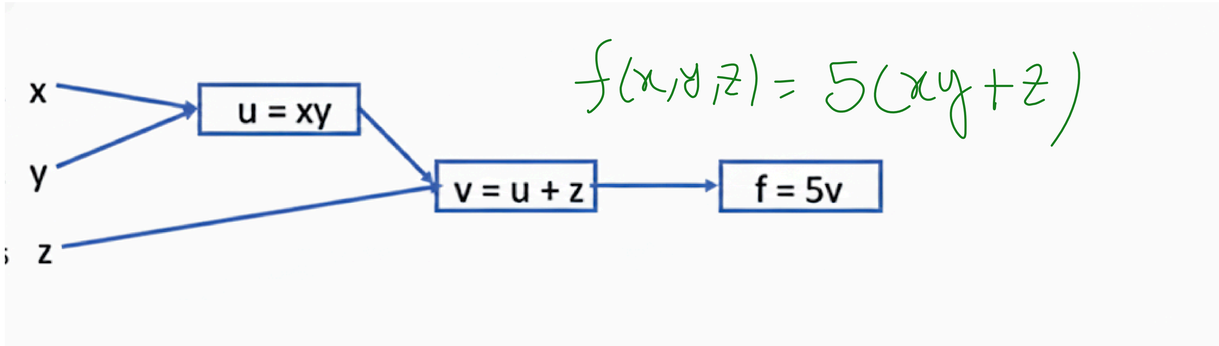
$$\frac{\partial c}{\partial a} = \frac{\partial c}{\partial b} \times \frac{\partial b}{\partial a} = 10 \times 2a = 2$$

$$200$$

$$V = uR$$

$$\frac{\partial I}{\partial a}$$

$$uR = V$$



$$f(x,y,z) = 5(xy+z)$$

Backpropagation

$$\left[\frac{\partial f}{\partial v}, \frac{\partial f}{\partial u}, \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right]$$

$$\frac{\sqrt{2}}{2}\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\right)$$